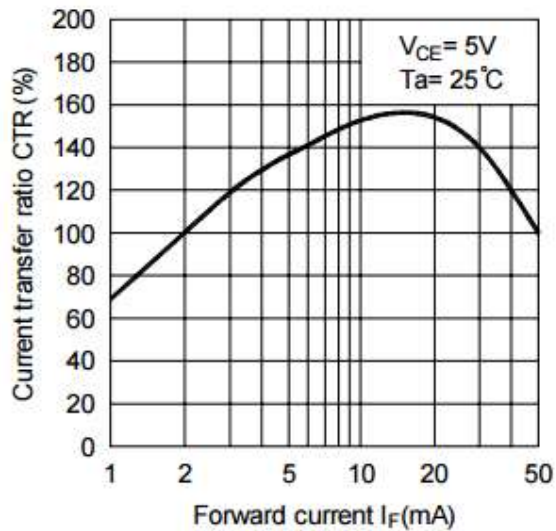


Fig.5 Current Transfer Ratio vs.
Forward Current



Electronics Optocoupler

Optocoupler Device CTR and Circuit CTR Explained

by [electronicsbeliever](#) • 0 • Electronics, Optocoupler

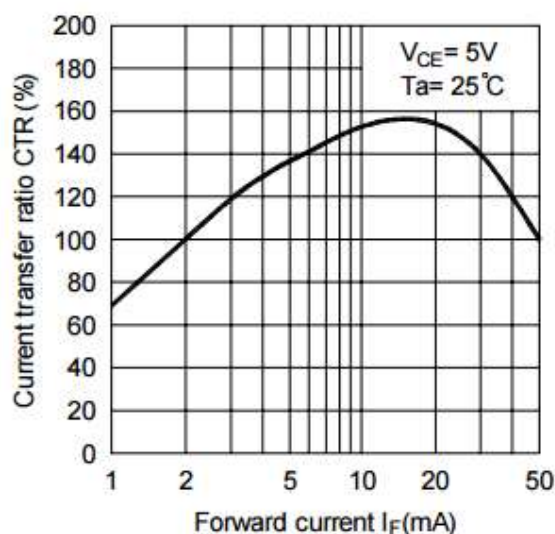
When you already heard about Optocoupler devices, you may be heard also about CTR. CTR stands for current transfer ratio. In bipolar junction transistor this is the so called current gain or simply beta. It is the ratio of the collector to forward current of an Optocoupler. In circuit design involving optocoupler, you may be dealing with the device CTR and circuit CTR. Don't be confused about these. Device CTR is the one specified in the component datasheet. Actually this is the CTR range that the device can still provide current transfer. In BJT, this is the range of beta wherein the device can still provide current amplification. In other words, this is the CTR that can make the circuit operate in linear mode.

If you check an Optocoupler datasheet, you can see a table like below. This is showing the CTR range of a particular device.

MODEL NO.	RANK MARK	CTR (%)
LTV-816	L	50 ~ 100
LTV-816	A	80 ~ 160

In my design approach I am not using the above table CTR range instead I look for the graph relating the forward current and the CTR as below.

Fig.5 Current Transfer Ratio vs.
Forward Current



The graph is more accurate than the table data above since it has forward current information. For example the computed forward current is 10mA, from the graph the equivalent CTR is around 155%.

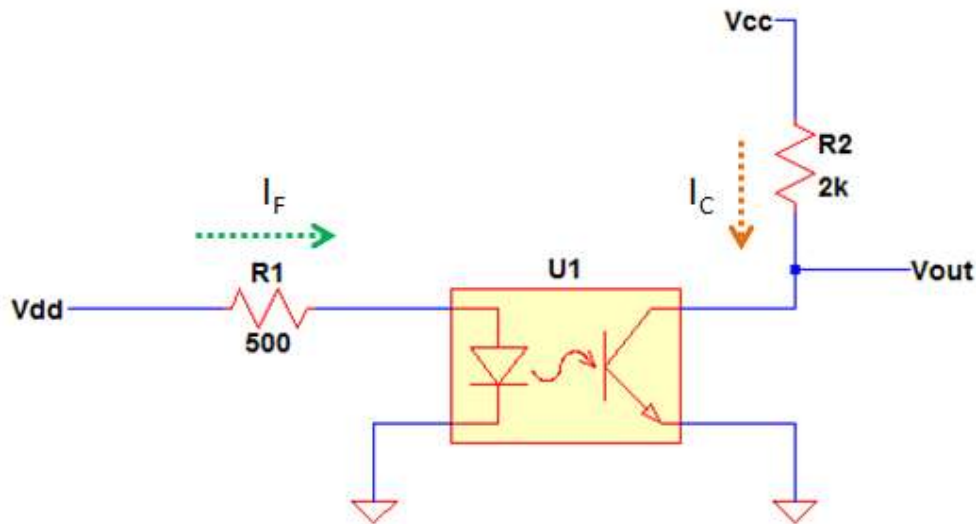
In linear operation, device CTR and circuit CTR is the same and can be expressed mathematically as below equation.

$$CTR = (I_c / I_f) \times 100\%$$

Go back to the graph above, any forward current value will have a corresponding CTR value and the collector current is can be solved as below equation.

$$I_c = CTR \times I_f$$

If we are going to relate in on a circuit like below, the value of I_c is dependent to the computed forward current and the corresponding CTR in the graph. This is always true as long as the operation of the circuit is at linear region.



However, when the above circuit saturates the collector current is only opposed by the value of the collector resistor R2. So the collector current is can be solved like below.

$$I_c = V_{cc} / R_2$$

At saturation, the method used above by getting the collector current based from the forward current and the CTR from the graph is not anymore valid. If you still use it, you will get higher collector current that results to a negative collector-emitter voltage.

The ratio of the collector current to the forward current computed at saturation is the one called as the circuit CTR.

To summarize everything, optocoupler device CTR and circuit CTR is the same as long as the device or circuit is operating in the linear region. Once the operation is entering saturation, optocoupler device CTR and circuit CTR circuit is no longer the same. The former is always higher than the latter. Device CTR is affected by by several factors such as forward current, operating temperature and life or aging. To know exactly how these factors affect optocoupler CTR, read [this](#).

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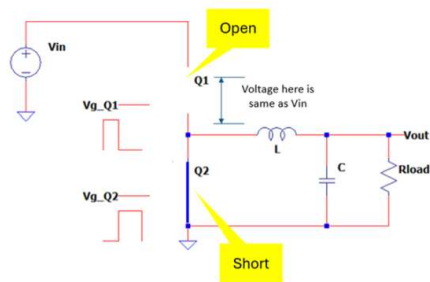
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Quick Facts About Synchronous Buck Converter

When you already heard about Optocoupler devices, you may be heard also about CTR. CTR stands for current transfer ratio. ... [Read more](#)

SUPPLY VOLTAGE		
Parameters	Value	Unit
V _{GG}	5	volt
V _{DD}	5	volt

This is the typical value of V_{GG}. See the figure.

This is the typical value of V_{DD}. See the figure.

AMBIENT TEMPERATURE		
Parameters	Value	Unit
T _{amb} max	40	°C
T _{amb} min	-30	°C

This is the maximum operating ambient temperature. This is a design given.

This is the minimum operating ambient temperature. This is a design given.

RESISTOR DATA		
Parameters	Value	Unit
R ₅₀₁	1500	ohms
R ₆₂	100000	ohms
is _{R1} rating	0.1	watt
is _{R2} rating	0.1	watt

Input here the resistance of R01. See figure.

Input here the resistance of R02. See figure.

Input here the power dissipation rating of R01. Get this from the datasheet.

Input here the power dissipation rating of R02. Get this from the datasheet.

RELAY DATA		
Parameters	Value	Unit
R _{coil} typical	100	ohms
L _{coil}	100	uH
is _{coil} rating	1	watt
is _{max} relay	40	°C
T _{max} relay	125	°C

Input here the coil resistance of the relay. Get this from the datasheet.

Input here the inductance of the relay. Get this from the datasheet.

Input here the power dissipation rating of the relay coil. Get this from the datasheet.

Input here the maximum operating temperature of the relay. Get this from the datasheet.

Input here the maximum operating temperature of the relay. Get this from the datasheet.

Design Template for NMOS Lowside Relay Driver

When you already heard about Optocoupler devices, you may be heard also about CTR. CTR stands for current transfer ratio. ... [Read more](#)



Buck Converter Design Using Excel Calculator

When you already heard about Optocoupler devices, you may be heard also about CTR. CTR stands for current transfer ratio. ... [Read more](#)



Synchronous Buck Converter Design Calculator Excel-External Switches

When you already heard about Optocoupler devices, you may be heard also about CTR. CTR stands for current transfer ratio. ... [Read more](#)

Parameter	Symbol	Value	Unit
Input Data			
Minimum input voltage	V_{inmin}	220	V
Maximum input voltage	V_{inmax}	240	V
Output voltage	V_o	400	V
Output current	I_o	400	mA
Normal output voltage	V_{on}	415	V
Normal output current	I_{on}	400	mA
Load Current			
Minimum output voltage (full load)	V_{oL}	24	V
Maximum output voltage (no load)	V_{oH}	24	V
Normal output voltage (nominal full load)	V_{on}	24	V
Normal output current (nominal full load)	I_{on}	24	A
Transformer Properties			
Resonance frequency	f_r	500	kHz
Resonance frequency	f_r	500	kHz
Quality factor	Q	10	
Output Power			
Minimum output power	P_{oL}	500	W
Maximum output power	P_{oH}	500	W
Normal output power	P_{on}	500	W
Conversion Rate			
Conversion rate efficiency	η	0.999999	
Conversion rate efficiency	η	0.9	
Conversion rate efficiency	η	0.999999999	
Minimum gain	G	0.999999999	

Template Legend

- Input signal: Input must provide
- Load signal: With formulas, do not alter
- Resonant capacitor
- Resonant inductor
- Transformer primary inductance

LLC AC Circuit

Half Bridge LLC Resonant Converter Power Section Calculator

When you already heard about Optocoupler devices, you may be heard also about CTR. CTR stands for current transfer ratio. ... [Read more](#)

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